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Power System Compensation

Grid-Connected STATCOM System: Reactive Power Compensation and Voltage Control

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Simulink Model Link:
Click here to access the Simulink Model on GitHub

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Abstract

This report presents the mathematical modeling, control design, and simulation of a grid-connected Static Synchronous Compensator (STATCOM) using MATLAB/Simulink. The STATCOM utilizes a Two-Level Voltage Source Converter (VSC) to provide dynamic reactive power compensation for a local inductive load, thereby improving the grid's power factor to unity. A Synchronous Reference Frame ($dq0$) control strategy is implemented. The outer loop regulates the DC-link voltage to a constant 800V, while the inner loops regulate the active and reactive current components. Sinusoidal Pulse Width Modulation (SPWM) with a 10kHz carrier frequency is employed to generate the converter's gate signals. Simulation results validate the STATCOM's ability to compensate 100 kVAR of reactive power, relieving the main grid from supplying inductive currents.

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1 System Specifications and Overview

The system models a localized industrial load supplied by a stiff AC grid. The load inherently consumes both active and reactive power, which drops the overall power factor and strains the transmission network. To mitigate this, a STATCOM is connected in parallel with the load to locally supply the required reactive power.

The grid operates at a nominal voltage of 415 V (line-to-line RMS) and a frequency of 50 Hz. The local load is a Three-Phase Parallel RLC branch consuming 200 kW of active power and 100 kVAR of inductive reactive power.

The STATCOM consists of a 10 mF DC-link capacitor, a Two-Level VSC, and an interfacing LCL filter (with 300 μH and 500 μH inductors and 100 μF capacitors) that smooths out the high-frequency switching ripples before injection into the grid. Table 1 summarizes the system's parameters.

Table 1: STATCOM System Simulation Parameters

Parameter	**Value**
AC Grid Voltage (V_{LL})	415 V (rms)
System Frequency (f)	50 Hz
Load Active Power (P_L)	200 kW
Load Reactive Power (Q_L)	100 kVAR (Inductive)
DC Link Target Voltage (V_{DC}^*)	800 V
DC Link Capacitor (C_{DC})	10 mF
PWM Carrier Frequency (f_{sw})	10 kHz

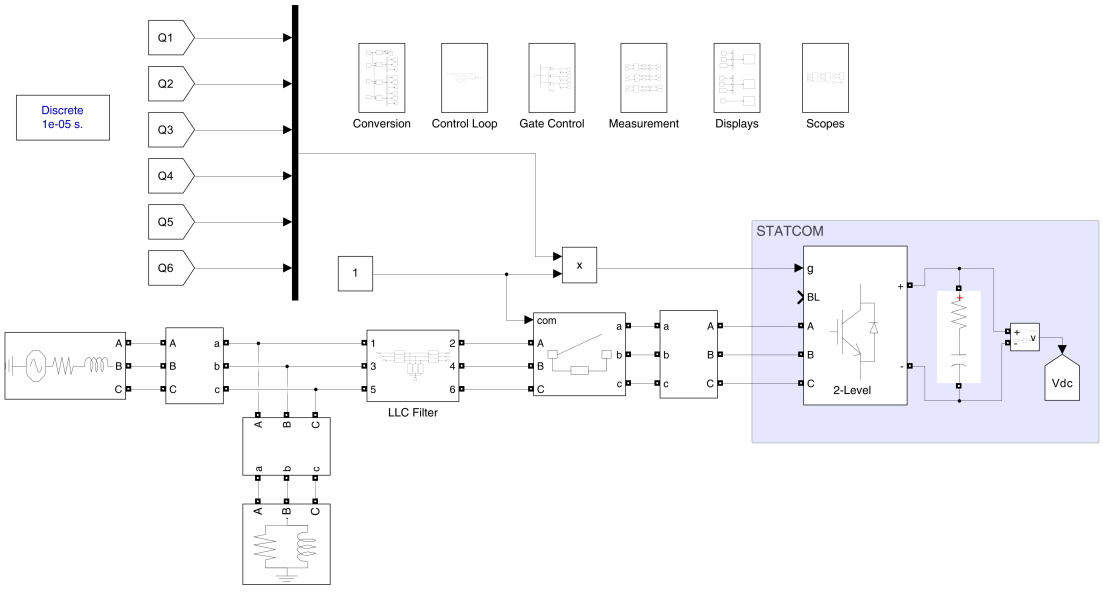


Figure 1: Overall Architecture of the Grid-Connected STATCOM System.

2 Control Strategy

The STATCOM's performance heavily relies on the Synchronous Reference Frame ($dq0$) control theory. This strategy transforms the three-phase time-varying AC signals (abc) into DC quantities ($dq0$) rotating at the synchronous speed, allowing the use of conventional Proportional-Integral (PI) controllers to achieve zero steady-state error.

2.1 Phase-Locked Loop (PLL) and Coordinate Transformation

A Three-Phase Phase-Locked Loop (PLL) continuously tracks the grid voltage to extract its phase angle (ωt). As illustrated in Figure 2, this angle is fed into the abc-to-$dq0$ transformation blocks (Park's Transformation) to decompose the grid voltages, load currents, and STATCOM currents into active (d-axis) and reactive (q-axis) components. This decoupling enables independent control over the active power (which dictates the DC-link voltage) and the reactive power (which dictates the compensation level).

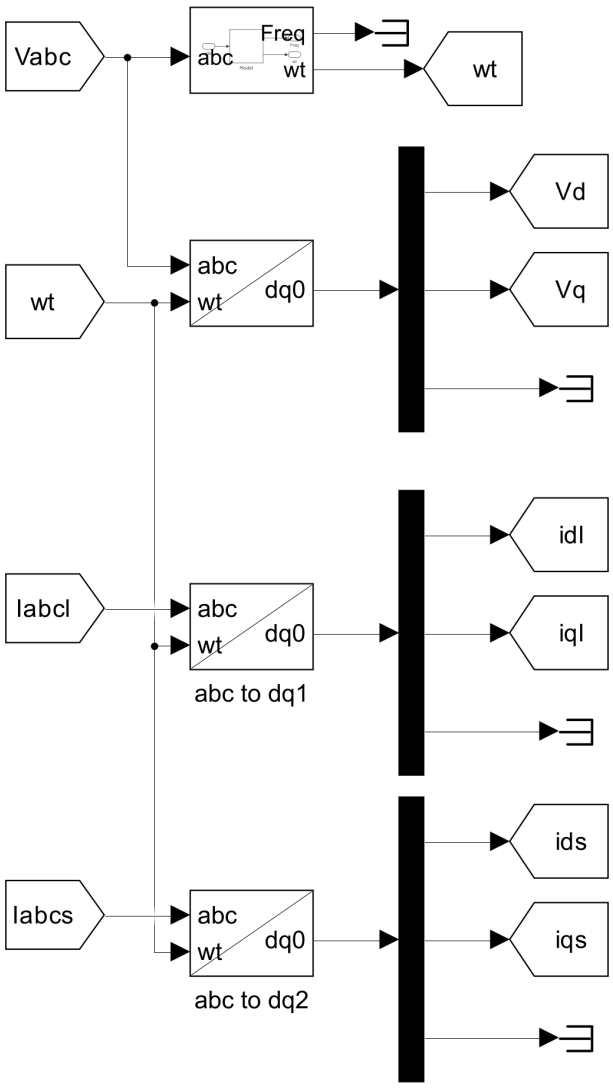


Figure 2: Coordinate Transformation Subsystem (abc to $dq0$).

2.2 Inner and Outer PI Control Loops

The control architecture is organized into cascaded loops, detailed in Figure 3:

- **Outer Voltage Loop:** Maintains the DC-link voltage at 800V. The error between the measured V_{DC} and the 800V reference is processed by a PI controller ($K_p = 5, K_i = 300$). The output of this loop represents the reference active current (I_d^*) required to replenish the converter's switching and conduction losses.
- **Inner Current Loops:** Two parallel PI controllers ($K_p = 25, K_i = 500$) regulate the d-axis and q-axis currents. The reactive current reference (I_q^*) is determined by the load's reactive power demand (100 kVAR) to ensure full compensation.

The outputs of the current controllers are transformed back to the abc frame using the inverse Park transformation ($dq0$-to-abc) to form the modulating reference signals (V_{ref}).

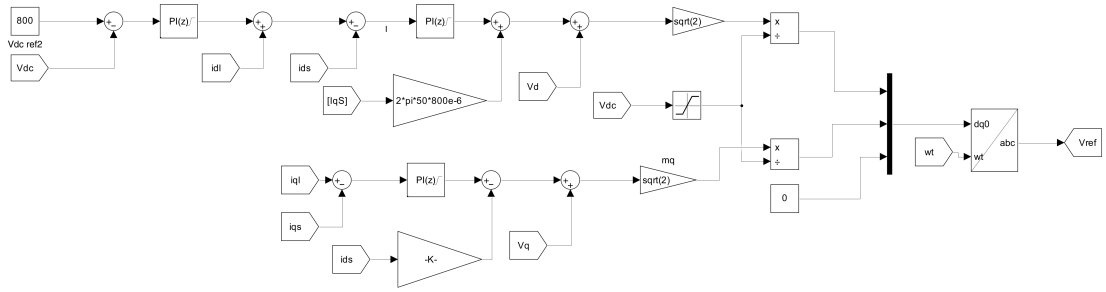


Figure 3: The $dq0$ Control Loop Subsystem showing PI controllers and Transformations.

2.3 Sinusoidal Pulse Width Modulation (SPWM)

The continuous reference signals (V_{ref}) generated by the control loop are fed into the Gate Control subsystem (Figure 4). Here, they are compared against a high-frequency triangular carrier wave using relational operators. The carrier wave is implemented using a Repeating Sequence block configured for a time period of $100\,\mu$s, which translates to a precise 10 kHz switching frequency. The logic produces the necessary complementary gate pulses (Q_1 to Q_6) for the 6 IGBTs of the Two-Level VSC.

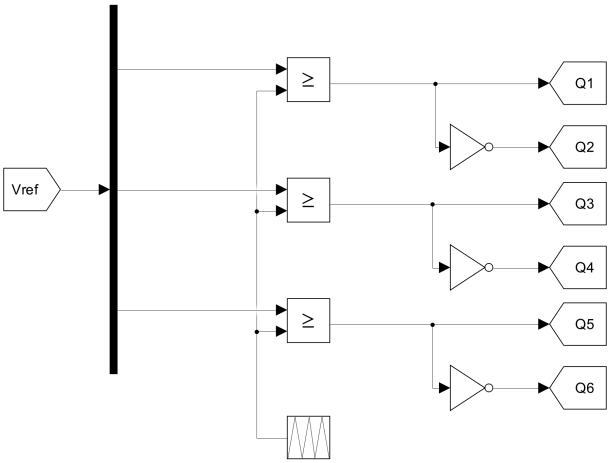
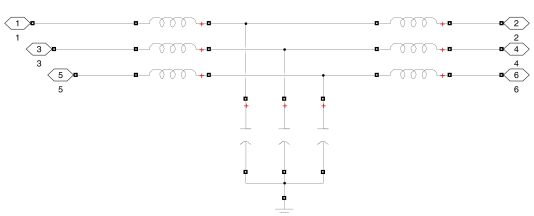


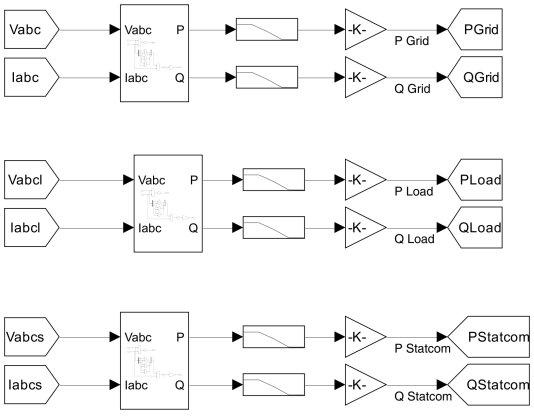
Figure 4: Gate Control Subsystem utilizing SPWM at 10 kHz.

2.4 LCL Filter and Measurements

To interface the VSC with the grid, a third-order LCL filter is employed (Figure 5). This filter configuration gives stronger attenuation than a simple L-filter and reduces the 10 kHz switching components entering the AC grid (a THD measurement against a standard is not part of this study). Furthermore, the Measurement subsystem (Figure 6) utilizes first-order low-pass filters to process the instantaneous active and reactive power signals for smooth monitoring and analysis.



(a) Three-Phase LCL Filter



(b) Measurement and Filtering Subsystem

Figure 5: Grid Interfacing Filter and Measurement Architecture.

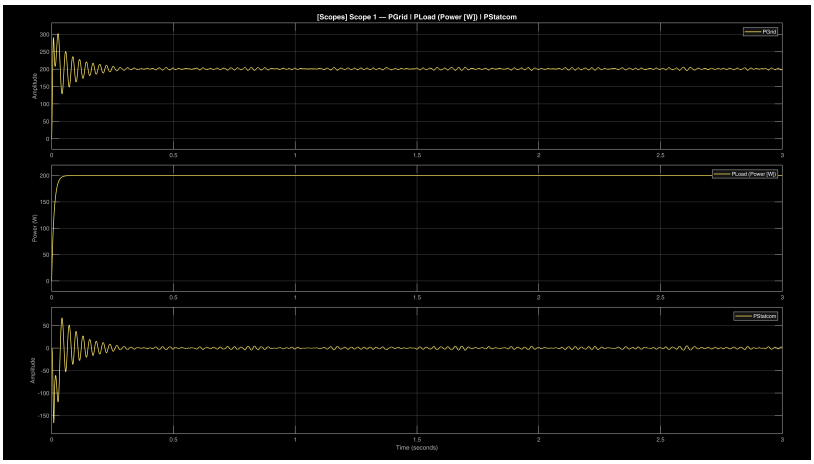
3 Simulation Results and Analysis

The simulation was executed to validate the STATCOM's ability to maintain a unity power factor at the grid connection point by dynamically supplying the required reactive power.

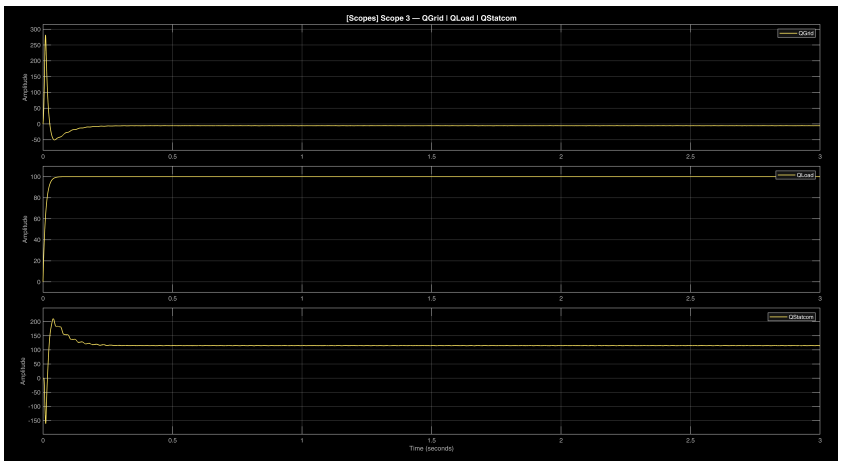
3.1 Active and Reactive Power Flow

The most significant result is the power flow distribution among the Grid, Load, and STATCOM. As shown in Figure 7, the local load continuously demands 200 kW and 100 kVAR. Once the STATCOM is engaged and reaches steady-state (after an initial transient period of approx. 0.2 s), it injects precisely 100 kVAR (capacitive) into the point of common coupling (PCC).

Consequently, the grid is relieved from supplying reactive power ($Q_{Grid} \approx 0$), ensuring a unity power factor and minimizing transmission losses. The active power supplied by the grid is approximately the 200 kW load demand plus a small margin to sustain the STATCOM's internal switching losses.



(a) Active Power Flow (P)

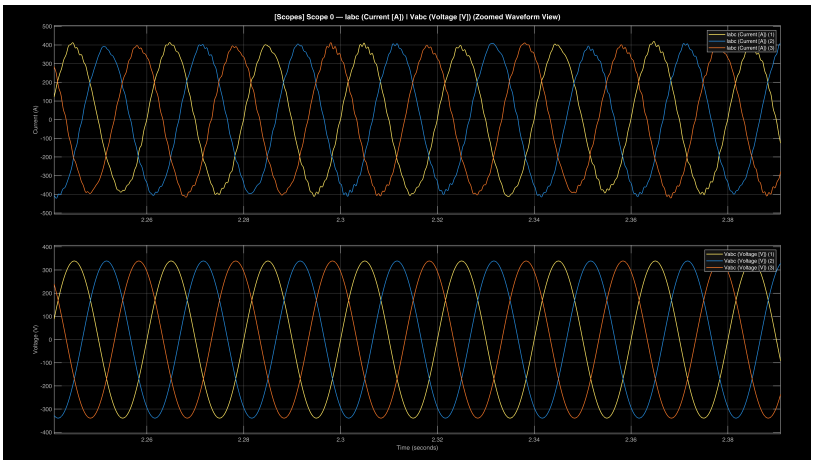


(b) Reactive Power Flow (Q)

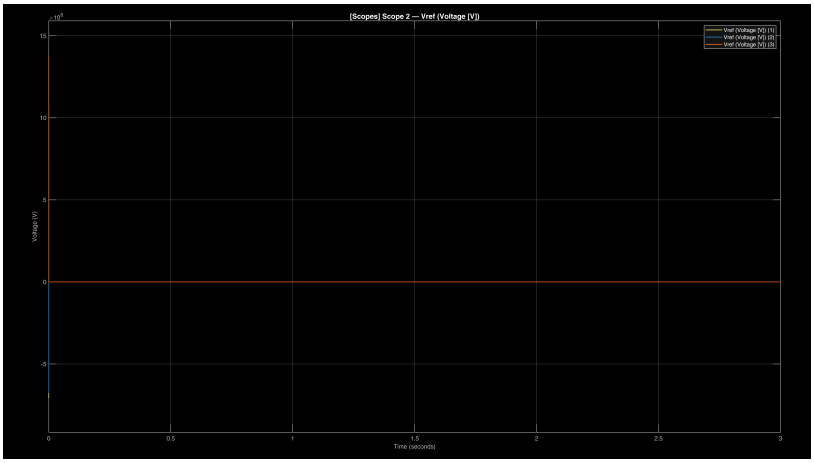
Figure 6: Power flow scopes demonstrating full reactive power compensation by the STATCOM.

3.2 Current and Voltage Waveforms

Figure 8 displays the dynamic response of the system currents and the generated reference voltage. The currents exhibit a transient behavior during the startup phase (0 to 0.2 s) as the DC-link charges and the PLL locks onto the grid frequency. After 0.2 s, the system enters a stable steady-state operation. The LCL filter effectively attenuates the 10 kHz switching harmonics, resulting in sinusoidal currents.



(a) Three-phase zoomed steady-state grid current and voltage waveforms (I_{abc} and V_{abc}).



(b) DC-link capacitor voltage regulation at nominal 800 V.

Figure 7: Dynamic response of system currents and voltage regulation.

4 Conclusion

The design and simulation of the grid-connected STATCOM were carried out. The implementation of the $dq0$ Synchronous Reference Frame control allowed for precise, decoupled regulation of the DC-link voltage and reactive power. The STATCOM compensated the 100 kVAR inductive load demand, achieving a unity power factor at the grid side. The PI controller tuning gave a stable 800 V DC-link, and the high-frequency SPWM coupled with the LCL filter gave sinusoidal grid currents. This project solidifies the importance of flexible AC transmission systems (FACTS) in enhancing modern grid stability.